

Research Notes on Sentiment analysis

Sentiment analysis

>> Section 1

The rise of social media such as blogs and social networks has fueled interest in sentiment analysis. With the proliferation of reviews, ratings, recommendations and other forms of online expression, online opinion has turned into a kind of virtual currency for businesses looking to market their products, identify new opportunities and manage their reputations. As businesses look to automate the process of filtering out the noise, understanding the conversations, identifying the relevant content and actioning it appropriately, many are now looking to the field of sentiment analysis. Further complicating the matter, is the rise of anonymous social media platforms such as 4chan and Reddit. If web 2.0 was all about democratizing publishing, then the next stage of the web may well be based on democratizing data mining of all the content that is getting published. One step towards this aim is accomplished in research. Several research teams in universities around the world currently focus on understanding the dynamics of sentiment in e-communities through sentiment analysis. The problem is that most sentiment analysis algorithms use simple terms to express sentiment about a product or service. However, cultural factors, linguistic nuances, and differing contexts make it extremely difficult to turn a string of written text into a simple pro or con sentiment. The fact that humans often disagree on the sentiment of text illustrates how big a task it is for computers to get this right. The shorter the string of text, the harder it becomes. Even though short text strings might be a problem, sentiment analysis within microblogging has shown that Twitter can be seen as a valid online indicator of political sentiment. Tweets' political sentiment demonstrates close correspondence to parties' and politicians' political positions, indicating that the content of Twitter messages plausibly reflects the offline political landscape. Furthermore, sentiment analysis on Twitter has also been shown to capture the public mood behind human reproduction cycles globally, as well as other problems of public-health relevance such as adverse drug reactions. While sentiment analysis has been popular for domains where authors express their opinion rather explicitly ("the movie is awesome"), such as social media and product reviews, only recently robust methods were devised for other domains where sentiment is strongly implicit or indirect. For example, in news articles - mostly due to the expected journalistic objectivity - journalists often describe actions or events rather than directly stating the polarity of a piece of information. Earlier approaches using dictionaries or shallow machine learning features were unable to catch the "meaning between the lines", but recently researchers have proposed a deep learning based approach and dataset that is able to analyze sentiment in

news articles. Scholars have utilized sentiment analysis to analyse construction health and safety posts on Twitter (which is called X now). The research revealed that there is a positive correlation between favorites and retweets in terms of sentiment valence. Others have examined the impact of YouTube on the dissemination of construction health and safety knowledge. They investigated how emotions influence users' behaviors in terms of viewing and commenting through semantic analysis. In another study, positive sentiment accounted for an overwhelming figure of 85% in knowledge sharing of construction safety and health via Instagram.

>> Section 2

Example of a subjective sentence: 'We Americans need to elect a president who is mature and who is able to make wise decisions.' This analysis is a classification problem. Each class's collections of words or phrase indicators are defined for to locate desirable patterns on unannotated text. For subjective expression, a different word list has been created. Lists of subjective indicators in words or phrases have been developed by multiple researchers in the linguist and natural language processing field states in Riloff et al. (2003). A dictionary of extraction rules has to be created for measuring given expressions. Over the years, in subjective detection, the features extraction progression from curating features by hand to automated features learning. At the moment, automated learning methods can further separate into supervised and unsupervised machine learning. Patterns extraction with machine learning process annotated and unannotated text have been explored extensively by academic researchers. However, researchers recognized several challenges in developing fixed sets of rules for expressions respectably. Much of the challenges in rule development stems from the nature of textual information. Six challenges have been recognized by several researchers: 1) metaphorical expressions, 2) discrepancies in writings, 3) context-sensitive, 4) represented words with fewer usages, 5) time-sensitive, and 6) ever-growing volume.

>> Section 3

Metaphorical expressions. The text contains metaphoric expression may impact on the performance on the extraction. Besides, metaphors take in different forms, which may have been contributed to the increase in detection. Discrepancies in writings. For the text obtained from the Internet, the discrepancies in the writing style of targeted text data involve distinct writing genres and styles. Context-sensitive. Classification may vary based on the subjectiveness or objectiveness of previous and following sentences. Time-sensitive attribute. The task is challenged by some textual data's time-sensitive attribute. If a group of

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researchers want to confirm a piece of fact in the news, they need a longer time for cross-validation, than the news becomes outdated. Cue words with fewer usages. Ever-growing volume. The task is also challenged by the sheer volume of textual data. The textual data's ever-growing nature makes the task overwhelmingly difficult for the researchers to complete the task on time. Previously, the research mainly focused on document level classification. However, classifying a document level suffers less accuracy, as an article may have diverse types of expressions involved. Researching evidence suggests a set of news articles that are expected to dominate by the objective expression, whereas the results show that it consisted of over 40% of subjective expression. To overcome those challenges, researchers conclude that classifier efficacy depends on the precisions of patterns learner. And the learner feeds with large volumes of annotated training data outperformed those trained on less comprehensive subjective features. However, one of the main obstacles to executing this type of work is to generate a big dataset of annotated sentences manually. The manual annotation method has been less favored than automatic learning for three reasons: